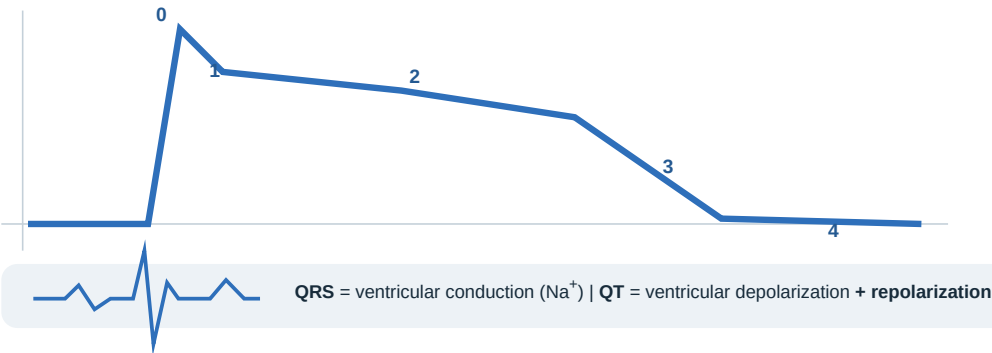


Build the electrical model first; then attach each drug to a tissue, interval, and use case.

1 ELECTRICAL MAP which phase, which tissue, which ECG interval?

FAST-RESPONSE CELL | atria, ventricles, Purkinje

QRS + QT TERRITORY

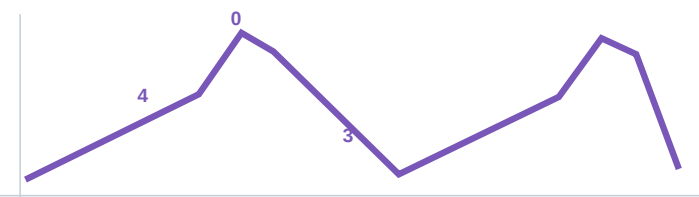


- 0 Na⁺ IN rapid upstroke / conduction CLASS I
- 2 Ca²⁺ IN plateau / contraction not the key Class IV site
- 3 K⁺ OUT repolarization / refractoriness CLASS III
- 4 K⁺ REST stable resting membrane substrate for next impulse

QRS = ventricular conduction (Na⁺) | QT = ventricular depolarization + repolarization

SLOW-RESPONSE CELL | SA + AV node

PR TERRITORY



Phase 4
spontaneous pacemaker slope
CLASS II flattens it

Phase 0
Ca²⁺-mediated upstroke
CLASS IV slows it

PR interval = AV-node delay. Longer PR = slower AV conduction.

2 HOW ARRHYTHMIAS START recognize the mechanism before choosing the tool

1 AUTOMATICITY

Phase-4 slope becomes steeper, threshold easier, or resting potential less negative.

Counter: reduce nodal drive / raise threshold.

2 RE-ENTRY | MOST COMMON

Requires a circuit, **unidirectional block**, and slow enough conduction for tissue to recover.

Counter: slow conduction or prolong ERP until the circuit fails.

3 EAD

Triggered during phase 2-3 when repolarization is prolonged; think **long QT**.

Risk: torsades de pointes.

4 DAD

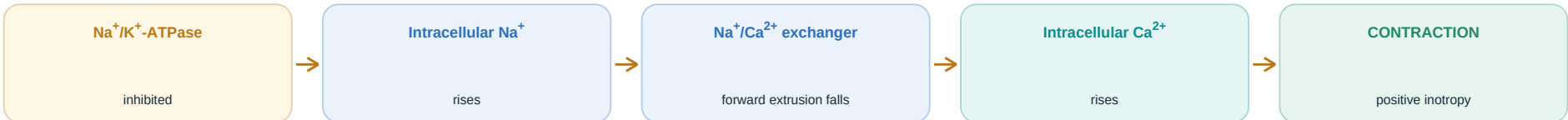
Triggered after repolarization by intracellular Ca²⁺ overload.

Think: digoxin toxicity / catecholamines.

3 LECTURE 18 DRUGS digoxin is the antiarrhythmic bridge; inotropes are kept separate

DIGOXIN | contraction up, AV conduction down

CLASSIC PAIR: HF + AF



VAGAL + ECG EFFECT: SA rate down + AV conduction down + AV refractory period up -> rate control; **PR lengthens**. Therapeutic pattern may show a **scooped ST segment + shorter QT**.

WHEN IT EARNS A PLACE

AF / flutter rate control, especially when heart failure or low blood pressure limits other AV-node blockers.

HF: improves symptoms / reduces hospitalization; **no survival benefit**.

TOXICITY PATTERN

GI: anorexia, nausea, vomiting
Neuro/visual: confusion, yellow vision / halos
Heart: almost any rhythm; classic clues are **atrial tachycardia + AV block** or bidirectional VT

RISK -> RESCUE

Risk: **renal failure, low K⁺/Mg²⁺, high Ca²⁺**, interactions.
1 Stop + ECG/K/Mg/Cr **2** Correct deficits / treat rhythm **3** Life-threatening -> **digoxin immune Fab**.

ACUTE INOTROPES | NOT Vaughan-Williams; proarrhythmic

MILRINONE / INAMRINONE

PDE3 block -> cAMP rises; **inodilator**.

USE: acute low-output HF; **AE**: hypotension + arrhythmia

DOBUTAMINE

Mainly beta1 agonist -> contractility / CO rise.

USE: low-output HF/shock; **AE**: tachyarrhythmia + ischemia

DOPAMINE

Dose-dependent beta1 / alpha effects.

USE: hypotensive shock; **AE**: arrhythmia/ischemia; no renal protection

GLUCAGON

Gs -> cAMP rises independent of beta receptor.

USE: severe beta-blocker overdose; **AE**: vomiting + hyperglycemia

USE-DEPENDENCE

Na⁺-channel blockers hit fastest in **tachycardic, depolarized, ischemic** tissue.

ECG LOCK

PR = AV node | QRS = ventricular conduction | QT = depolarization + repolarization.

UNSTABLE RHYTHM

Hypotension, shock, ischemic chest pain, acute HF, or altered mental status -> **electricity first**.

ANTIARRHYTHMICS | CLASS -> DRUG -> PRESENTATION -> NEXT STEP

Core spine: I = fast Na+ | II = beta | III = K+ | IV = non-DHP Ca2+

1 THE FOUR-CLASS SPINE mechanism, ECG change and main clinical job

I FAST SODIUM	II BETA BLOCKERS	III POTASSIUM	IV CALCIUM
Na+ BLOCK IA also K+	BETA1 -> cAMP DOWN	K+ BLOCK	non-DHP L-TYPE Ca2+
WHERE: atria, ventricles and Purkinje MECH: fast Na+ block -> phase-0 upstroke and conduction slow ECG: QRS widens (IC > IA > IB); IA also lengthens QT BEST USE: atrial + ventricular rhythm control	WHERE: SA and AV nodes MECH: funny current/Ca entry fall -> phase-4 slope and AV conduction slow ECG: heart rate falls; PR lengthens BEST USE: AF rate, AVNRT, catecholamine rhythms, post-MI	WHERE: working atrial + ventricular myocardium MECH: phase-3 repolarization slows -> APD and ERP lengthen ECG: QT lengthens BEST USE: AF rhythm control + ventricular arrhythmias	WHERE: SA and AV nodes MECH: nodal phase-0 upstroke and AV conduction slow ECG: PR lengthens BEST USE: AVNRT + AF/flutter rate control

2 DRUGS + ADVERSE-EFFECT LOCKS each card answers drug, use, toxicity and avoid

CLASS I fast Na+ blockers	AV NODE + OTHER	CLASS III K+ block
QRS: IC > IA > IB		QT UP / TORSADES
IA moderate Na + K block DRUGS: quinidine procainamide disopyramide USE: Procainamide for stable pre-excited AF / selected stable VT; others now uncommon. AE / AVOID: Torsades. Quinidine - cinchonism, diarrhea, thrombocytopenia; procainamide - lupus, agranulocytosis; disopyramide - antimuscarinic effects + marked negative inotropy.	II beta blockers DRUGS: metoprolol esmolol propranolol USE: AF/flutter rate control, AVNRT, catecholamine rhythms and post-MI protection. AE / AVOID: Bradycardia, AV block, hypotension, fatigue; nonselective agents -> bronchospasm. Caution in acute decompensated HF.	AMIODARONE multichannel USE: AF rhythm + VT/VF; established HFrEF option AE: lung, thyroid, liver, cornea/skin, neuropathy, bradycardia, interactions; very long half-life. QT up but comparatively low torsades.
IB weak / fastest off DRUGS: lidocaine mexiletine USE: Ischemic or digoxin-triggered ventricular arrhythmias; lidocaine acute, mexiletine oral. AE / AVOID: CNS toxicity, tremor or seizures (lidocaine); GI + CNS effects (mexiletine). Little atrial effect.	IV non-DHP Ca blockers DRUGS: verapamil diltiazem USE: AVNRT and AF/flutter rate control when systolic function permits. AE / AVOID: Bradycardia, AV block, hypotension, edema; verapamil -> constipation. Avoid HFrEF.	DOFETILIDE pure IKr USE: AF conversion/maintenance; established HFrEF option AE: QT/torsades + renal dosing; initiate or reload with inpatient continuous ECG.
IC strongest / slowest off DRUGS: flecainide propafenone USE: AF/SVT rhythm control only when there is no prior MI or important structural disease. AE / AVOID: QRS widening, 1:1 flutter, ventricular proarrhythmia, worsened HF; propafenone also causes beta-blocker effects/bronchospasm.	ADENOSINE SVT; flushing/chest pressure, AV block, bronchospasm	SOTALOL K block + beta block USE: AF maintenance + selected ventricular rhythms AE: Torsades plus bradycardia/bronchospasm; renal dosing. Generally avoid significant HFrEF.
	DIGOXIN AF + HF; GI/yellow vision, AV block or ectopy	DRONEDARONE no iodine USE: AF maintenance in selected patients AE: Liver injury, bradycardia, QT. Avoid permanent AF and recent decompensated/severe HF.
	MAGNESIUM torsades; flushing, hypotension; renal accumulation	ATROPINE symptomatic brady; tachycardia + antimuscarinic effects

3 PRESENTATION -> ESCALATION OR means equivalent alternatives; instability overrides every ladder

UNSTABLE = ELECTRICITY FIRST

PRESENTATION	1 START / PREFERRED	2 NEXT / ALTERNATIVE	3 RESCUE / OVERRIDE
REGULAR NARROW SVT Stable and usually AV-node dependent	Vagal maneuvers	IV adenosine	IV diltiazem/verapamil or beta blocker. Unstable or refractory -> synchronized cardioversion.
AF / FLUTTER WITH RVR First rule out pre-excitation	No pre-excitation: beta blocker OR diltiazem/verapamil if EF >40%	Digoxin when HFrEF or usual agents are limited; slow acute onset	Critically ill: IV amiodarone may be considered. Unstable/ischemia -> synchronized cardioversion.
AF RHYTHM CONTROL Selection is substrate-driven, not a fixed ladder	Normal heart: flecainide OR propafenone + AV-node blocker; dronedarone is another option	HFrEF: amiodarone OR dofetilide . QT, renal function and HF status guide Class III choice	Persistent symptoms or drug unsuitable -> catheter ablation . Avoid Class IC after MI/structural scar.
STABLE MONOMORPHIC WCT Treat uncertain wide tachycardia as VT	Monitor + IV + 12-lead + expert help	Procainamide OR amiodarone OR sotalol . Adenosine only if regular + monomorphic	If ineffective/contraindicated -> synchronized cardioversion. Never diltiazem/verapamil in WCT.
VF / PULSELESS VT Shockable cardiac arrest	Shock -> CPR	Next shock cycle: epinephrine during CPR	Further shock cycle: amiodarone OR lidocaine + treat reversible causes.
POLYMORPHIC VT / TORSADES Sustained polymorphic VT is unstable	Immediate unsynchronized shock	Recurrent long-QT torsades: IV magnesium + correct K⁺/Mg²⁺ + stop QT drugs	Pause-triggered recurrence: expert pacing/isoproterenol. Normal QT? Treat ischemia; Mg is not routine.
SYMPTOMATIC BRADYCARDIA Poor perfusion: hypotension, shock, ischemia, HF, altered mentation	Treat cause + atropine	If ineffective: transcutaneous pacing and/or dopamine or epinephrine infusion	Expert consultation -> transvenous pacing; check hypoxia, ischemia, drugs and hyperkalemia.



PRE-EXCITED AF / WPW **UNSTABLE -> synchronized cardioversion | STABLE -> IV procainamide OR ibutilide | DEFINITIVE -> accessory-pathway ablation.**

NEVER AV-NODE BLOCKERS: adenosine, beta blockers, diltiazem/verapamil, digoxin, or IV amiodarone - they can accelerate accessory-pathway conduction and trigger VF.